

A heterogeneous network-based method with attentive meta-path extraction for predicting drug–target interactions

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Abstract

Predicting drug–target interactions (DTIs) is crucial at many phases of drug discovery and repositioning. Many computational methods based on heterogeneous networks (HNs) have proved their potential to predict DTIs by capturing extensive biological knowledge and semantic information from meta-paths. However, existing methods manually customize meta-paths, which is overly dependent on some specific expertise. Such strategy heavily limits the scalability and flexibility of these models, and even affects their predictive performance. To alleviate this limitation, we propose a novel HN-based method with attentive meta-path extraction for DTI prediction, named HampDTI, which is capable of automatically extracting useful meta-paths through a learnable attention mechanism instead of pre-definition based on domain knowledge. Specifically, by scoring multi-hop connections across various relations in the HN with each relation assigned an attention weight, HampDTI constructs a new trainable graph structure, called meta-path graph. Such meta-path graph implicitly measures the importance of every possible meta-path between drugs and targets. To enable HampDTI to extract more diverse meta-paths, we adopt a multi-channel mechanism to generate multiple meta-path graphs. Then, a graph neural network is deployed on the generated meta-path graphs to yield the multi-channel embeddings of drugs and targets. Finally, HampDTI fuses all embeddings from different channels for predicting DTIs. The meta-path graphs are optimized along with the model training such that HampDTI can adaptively extract valuable meta-paths for DTI prediction. The experiments on benchmark datasets not only show the superiority of HampDTI in DTI prediction over several baseline methods, but also, more importantly, demonstrate the effectiveness of the model discovering important meta-paths.

Keywords: drug–target interaction, heterogeneous graph, meta-path, graph neural networks

Introduction

The process of drug discovery, where medical researchers devote much effort to screening candidate molecules from vast chemical space, suffers from prohibitive cost and extreme time consumption [1]. As a remedy, drug repositioning aiming at identifying unintended and potential targets for drugs can reduce the time and cost required for drug discovery to a certain extent [2, 3]. In the current era of big data, computationally predicting drug–target interactions (DTIs) promises to guide *in vivo* or wet-lab validation and significantly accelerate drug discovery or repositioning [4, 5]. Early computational methods that mainly depend on two strategies, the docking-based methods [6] and the ligand-based methods [7], are limited in prediction accuracy. Ligand-based methods perform poorly when the target protein has few known binding ligands, and the docking-based methods overly dependent on the 3D structures of target proteins often produce unsatisfactory prediction results when the target structure information is inaccessible [2].

A favorite modern solution drawing much attention is machine learning-based DTI prediction, which has been

rapidly developing in the past decade [8]. The majority of machine learning-based models for DTI prediction are based on the intuition that similar drugs may share similar targets and *vice versa*, so-called ‘guilt-by-association’ assumption. In general, there are many factors that affect how a drug interacts with a target in complex biology systems. Thereby, those machine learning-based models built on one single data source fail to sufficiently capture the complexity of molecule machinery with which DTIs associate [9] and produce limited predictive performances. With the continuous accumulation of massive genomic, biochemical and biomedical data, fusing heterogeneous data sources that is believed to enable more accurate DTI prediction has become the major concern in recent years [10–12]. Among these advances, the computational models and intelligent algorithms for integrating the relevant heterogeneous networks (HNs) offer a more rational roadmap for DTI prediction, as the complicated connectivity and non-linear structure of the HN make it more appropriate for modeling the integrative context fraught with DTI-related complex pharmacology and biology [9, 13]. Some shallow models

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dissecting HNs for DTI prediction, such as kernel learning [14, 15], label propagation [16, 17] and combining network embedding models with an external predictor [2, 12, 18], have shown promises in predictive performance. More recently, deeper models [19–22] that can effectively capture complex topological structure of HNs, including various graph neural networks (GNNs) [10, 23–26], have further boosted the accuracy of HN-based DTI prediction.

A common approach to dealing with HNs is based on meta-path. The meta-path is a sequence of node types connected by relation types in the HN and contains rich semantic information. In biological HNs, meta-paths represent some metabolic pathways or biology principles. The analysis of meta-paths in an HN involving drugs and targets could make great sense for explainable DTI prediction. Fu et al. [27] and Thafar et al. [28] scored meta-paths and fed the path scores into a classifier to predict DTIs. Chen et al. [26] and Li et al. [25] devised meta-path-based neighbor aggregation within the respective proposed heterogeneous GNN to integrate semantic information of meta-paths. Wang et al. [22] used random walk guided by meta-paths to generate path sequences as the input of the follow-up model for predicting bio-entity relationships including DTIs. These methods have demonstrated the effectiveness of meta-paths for DTI prediction and especially the nearest deep models with analyzing meta-paths have shown the superiority in performance. However, selecting meta-paths beneficial to DTI prediction has not been well addressed. This problem is far-reaching, since the predictive performance dramatically shifts with the choice of meta-paths. For example, Li et al. [25] compared the results of six groups of meta-paths and found the performance differs by almost 20%. Moreover, this problem is challenging, because examining all meta-path combinations by brute-force enumeration is hardly feasible. But existing studies overlook this problem, where the meta-paths are usually customized by empirical selection. Such strategy is overly dependent on domain knowledge and not always adaptable and transferable across different HNs. Consequently, it significantly limits the scalability and flexibility of those methods. Besides, the manually defined meta-paths may contain redundancy and noise, which has impact on the performance of those models in predicting DTIs.

In order to mitigate the limitation, we propose a novel HN-based method with attentive meta-path extraction for DTI prediction, or HampDTI for short, which can automatically extract meaningful meta-paths without domain knowledge. Concretely, by scoring multi-hop connections across various relations in the HN with each relation assigned an attention weight, HampDTI constructs a new trainable graph structure, called meta-path graph. Such meta-path graph implicitly measures the importance of every possible meta-path between drugs and targets. To enable HampDTI to extract diverse meta-paths, we adopt a multi-channel mechanism to generate multiple meta-path graphs. Then, a GNN is deployed on the generated meta-path graphs to learn

the multi-channel embeddings of drugs and targets. Finally, HampDTI fuses all embeddings from different channels for predicting DTIs. The meta-path graphs are optimized in the training stage such that HampDTI can adaptively extract valuable meta-paths for DTI prediction. The experimental results show that HampDTI can outperform several baseline methods that were designed for HN-based DTI prediction. Moreover, we discuss the HampDTI's capability of extracting useful meta-paths which also provides good interpretability for meta-path-based DTI prediction.

Methods

Preliminaries

Definition 2.1 Heterogeneous Network. An HN with regard to DTIs is defined as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with multiple types of nodes and edges, where $\mathcal{V}$ is the node set and $\mathcal{E}$ is the edge set. Let $\mathcal{N}$ denote the associated node type set that includes drug and target or other bio-entities if they exist, and $\mathcal{R}$ denote the associated relation set that includes DTI or other bio-entity relations if they exist.

Definition 2.2 Meta-path. A meta-path is a sequence of node types linked by relation types. Formally, in the HN $\mathcal{G}$, a meta-path $\mathcal{M}$ with length of L can be represented as $n_1 \xrightarrow{r_1} n_2 \xrightarrow{r_2} n_3 \dots \xrightarrow{r_L} n_{L+1}$, where $r_i \in \mathcal{R}$ and $n_i \in \mathcal{N}$.

Definition 2.3 Meta-path Graph. A meta-path graph represents the connectivity between the nodes from the starting node type n_1 and ones from the ending node type n_{L+1} in the given meta-path $\mathcal{M}$.

Given a set of drugs $\mathcal{D} = \{d_1, d_2, \dots, d_M\}$ and a set of targets $\mathcal{T} = \{t_1, t_2, \dots, t_N\}$, the observed DTIs can be transformed into a binary matrix $\mathbf{Y} \in \{0, 1\}^{M \times N}$, where $\mathbf{Y}_{ij} = 1$ if a drug d_i is known to interact with a target t_j and $\mathbf{Y}_{ij} = 0$ indicates that the interaction between the drug d_i and the target t_j is unknown or unobserved. We formulate the HN-based DTI prediction as a matrix completion, the goal of which is to fill in the unknown entries of the matrix $\mathbf{Y}$ based on a relevant HN.

The HampDTI Method

HampDTI extracts useful meta-paths for DTI prediction by optimizing meta-path graph structure that implicitly represents the importance of all possible meta-paths. First, the **molecule characterizer** characterizes drugs and targets from drug molecule graphs and protein sequences, respectively. Meanwhile, the **meta-path graph constructor** constructs meta-path graph based on relation-specific attention. Then, the **GNN-based encoder** deployed on the constructed meta-path graph encodes the drug/target features output from the molecule characterizer into the embeddings of drugs and targets. To enable diverse meta-paths, we use different attention to construct different meta-path graphs and a shared GNN-based encoder to obtain multi-channel embeddings. Finally, the **DTI predictor** incorporates the multi-channel embeddings and reconstructs the DTI

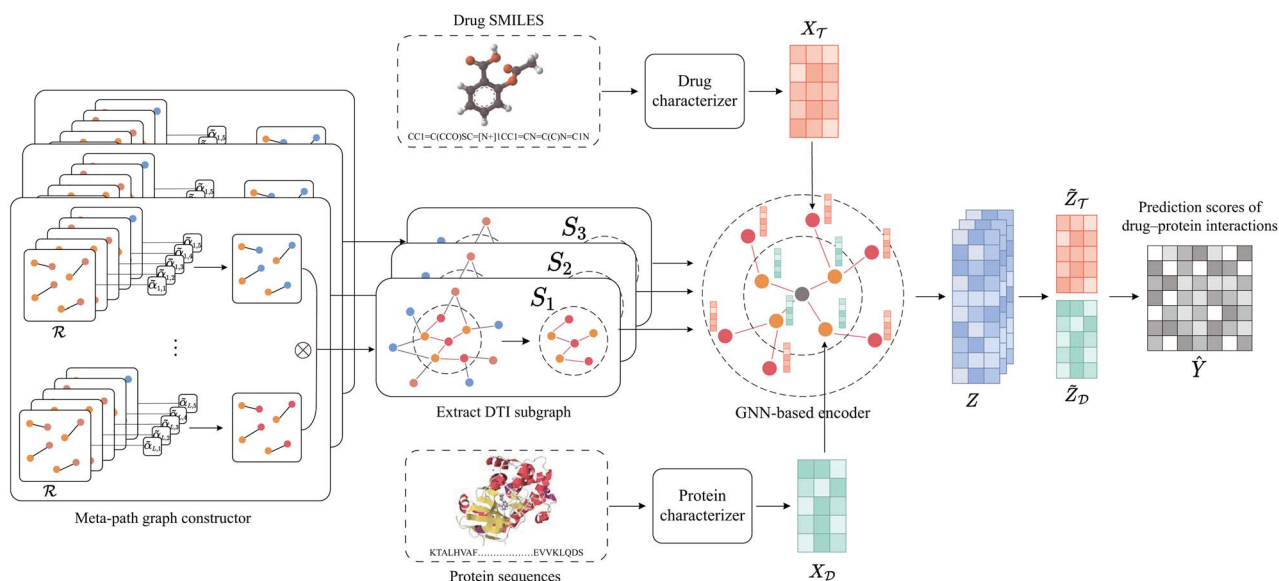


Figure 1. The schematic workflow of the HampDTI. HampDTI constructs multi-channel meta-path graphs and each channel is built by scoring multi-hop connections across the HN where each relation is assigned an attention weight. The node type-aware GNN deployed on the constructed meta-path graphs with node features learned by the molecule characterizer learns the embeddings of drugs and targets for predicting DTIs.

matrix $\mathbf{Y}$ to predict DTIs. Figure 1 illustrates the workflow of the HampDTI.

Molecule characterizer

Drug SMILES, which is a kind of description for the drug molecule, can be converted into the molecule graph by Rdkit [29]. In a molecule graph, the nodes represent atoms and the edges serve as chemical bonds. We adopt the vanilla graph convolutional network (GCN) [30] to extract drug features from drug molecule graphs. Formally, for a drug $d_i \in \mathcal{D}$, the atom embeddings at the l^{th} layer are defined as:

$$\mathbf{H}_i^{(l)} = \sigma \left(\tilde{\mathbf{D}}_i^{-\frac{1}{2}} \tilde{\mathbf{A}}_i \tilde{\mathbf{D}}_i^{-\frac{1}{2}} \mathbf{H}_i^{(l-1)} \mathbf{W}^{(l)} \right), \quad (1)$$

where $\tilde{\mathbf{A}}_i = \mathbf{A}_i + \mathbf{I}$ is the adjacency matrix with self-connections regarding the molecule graph of drug d_i , $\tilde{\mathbf{D}}_i$ is the degree matrix of $\tilde{\mathbf{A}}_i$, $\mathbf{W}^{(l)}$ is a weight matrix and σ is a non-linear activation function. Following the previous study [31], we use 78-dimensional atomic attributions as the initial atom embeddings of the drug d_i , i.e. $\mathbf{H}_i^{(0)} \in \mathbb{R}^{Q_i \times 78}$, where Q_i denotes the number of atoms contained in d_i . Followed by a single-layer GCN, a global max pooling layer is utilized to obtain the drug feature of d_i , i.e. $\mathbf{x}_i \in \mathbb{R}^F = \text{pool}(\mathbf{H}_i^{(1)})$, where F is the embedding dimensionality. We compile all M drug features into a feature matrix $\mathbf{X}_{\mathcal{D}} \in \mathbb{R}^{M \times F}$.

For one target molecule $t_i \in \mathcal{T}$, its protein sequence can be encoded into a 147-dimensional vector via the Composition, Transition and Distribution (CTD) rule [32, 33]. Then, all the CTD representations are fed into a multi-layer perceptron (MLP) to obtain the compact protein features $\mathbf{X}_{\mathcal{T}} \in \mathbb{R}^{N \times F}$.

Meta-path graph constructor

In this section, we construct trainable meta-path graphs connecting drugs with targets based on the similar strategy from GTN [34]. According to Definition 2.2 and 2.3, meta-path graph is a graph structure that represents multi-hop connectivity in the HN where each hop/step corresponds to a relation type. We set the adjacent matrix of the HN $\mathcal{G}$ as $\mathbf{T} = \sum_{r \in \mathcal{R}} \mathbf{R}_r$. Here, the $\mathbf{R}_r \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$ is an adjacent matrix of the relation r and $|\mathcal{V}|$ is the number of nodes in $\mathcal{G}$. Note that the non-zero entry in $\mathbf{R}_r$ indicates the corresponding node pair is linked by the relation r . Simply, for a given meta-path $n_1 \xrightarrow{r_1} n_2 \xrightarrow{r_2} n_3 \cdots \xrightarrow{r_L} n_{L+1}$, we can measure the L -hop connectivity through the meta-path in $\mathcal{G}$ by the product of a sequence relation matrix as $\prod_{i=1}^L \mathbf{R}_{r_i}$.

For adaptive selection of meta-paths, we allocate specific weight for each relation called relation-specific attention and use different sets of attention weights at each hop, which can be formulated as:

$$\tilde{\mathbf{T}} = \prod_{i=1}^L \left(\sum_{r \in \mathcal{R}} \tilde{\alpha}_{i,r} \mathbf{R}_r \right), \text{ where } \tilde{\alpha}_{i,r} = \frac{\exp(\alpha_{i,r})}{\sum_{r' \in \mathcal{R}} \exp(\alpha_{i,r'})}. \quad (2)$$

Here, $\alpha_{i,r}$ is a learnable parameter and $\tilde{\alpha}_{i,r}$ represents the relation-specific attention weight. The attention $\tilde{\alpha}_{i,r}$ can also be viewed as the probability that relation r is selected as the i^{th} component of a latent meta-path with a given length L . Hence, $\tilde{\mathbf{T}}$ corresponds to a trainable graph structure that implies all possible L -length meta-paths in the HN. The well-trained attention weights will serve as the basis of prioritizing possible meta-paths which will be discussed in the Discussion section. Note that the constructed graph $\tilde{\mathbf{T}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ contains all nodes

in the HN; we only care about the meta-paths between drug and target, and hence extract DTI subgraph from $\tilde{\mathbf{T}}$ denoted as $\mathbf{S} \in \mathbb{R}^{(M+N) \times (M+N)}$ that is the desired meta-path graph.

In order to enable more diverse meta-paths, we consider two tricks: (1) we replace the relation set $\mathcal{R}$ in Eq. (2) with the augmented one $\mathcal{R} \cup \{\text{identical}\}$ to cover the meta-paths with length less than L , where the adjacent matrix of the relation *identical* is the identity matrix $\mathbf{I}$; (2) we repeat the graph construction C times with unshared attention for multi-channel design.

GNN-based encoder

For the goal of DTI prediction, we devise a node type-aware GNN to more deeply extract latent representations for drugs and targets. Concretely, for a constructed DTI graph $\mathbf{S}$, we consider arranging distinctive transformation function for drugs and targets within a K -iteration SGC [35], a simple graph convolution for avoiding over-complication of the model, which is formulated as:

$$\mathbf{z} = \tilde{\mathbf{S}}^K \begin{bmatrix} f_D(\mathbf{X}_D) \\ f_T(\mathbf{X}_T) \end{bmatrix}, \quad (3)$$

where $\mathbf{Z}$ is the node embeddings of drugs and targets, $\tilde{\mathbf{S}} = (\mathbf{D}_S + \mathbf{I})^{-\frac{1}{2}} (\mathbf{S} + \mathbf{I}) (\mathbf{D}_S + \mathbf{I})^{-\frac{1}{2}}$ is the renormalized convolution matrix, $\mathbf{D}_S$ is a diagonal matrix with the sum of each row of $\mathbf{S} + \mathbf{I}$ as the corresponding entry, f_D and f_T are two different MLPs for making the model aware the node type information and $\mathbf{X}_D$ and $\mathbf{X}_T$ are the drug features and target features described above.

Since we implement multi-channel design, all channels share the same GNN encoder and we can obtain multi-channel embeddings for drugs and targets. In addition, we consider learning an additional channel based on the observed DTI network. Hence, together with constructed graph of C channels, we obtain $C + 1$ embeddings for each drug or target node.

DTI predictor

Given that we have obtained multi-channel embeddings for drugs and targets, we merge them through an channel attention as follows:

$$\tilde{\mathbf{z}} = \sum_{i=1}^{C+1} \frac{\exp(\beta_i)}{\sum_{j=1}^{C+1} \exp(\beta_j)} \mathbf{z}_i, \quad (4)$$

where β_i is a learnable parameter and $\mathbf{z}_i$ is the i th channel of embeddings learned above. $\tilde{\mathbf{Z}} \in \mathbb{R}^{(M+N) \times F}$ is the concatenation of drug embeddings and target embeddings, and we detach it to obtain final drug representations $\tilde{\mathbf{Z}}_D \in \mathbb{R}^{M \times F}$ and target representations $\tilde{\mathbf{Z}}_T \in \mathbb{R}^{N \times F}$.

As mentioned before, the DTI prediction task is viewed as a matrix completion. Accordingly, we first reconstruct the DTI matrix by

$$\hat{\mathbf{Y}} = \tilde{\mathbf{Z}}_D \tilde{\mathbf{Z}}_T^T. \quad (5)$$

Table 1. Statistics of the DTInet dataset

Node types	Num	Edge types	Num
Drug	708	Drug-protein	1923
Protein	1512	Drug-drug	10 036
Disease	5603	Protein-protein	7363
Side Effect	4192	Drug-disease	199 214
		Protein-disease	1596 745
		Drug side effect	80 164
Total	12 015	Total	1895 445

Then, the HampDTI model can be optimized through the following loss function:

$$\ell = (1 - \gamma) \|\mathbf{Y} \odot (\mathbf{Y} - \hat{\mathbf{Y}})\|^2 + \gamma \|(1 - \mathbf{Y}) \odot (\mathbf{Y} - \hat{\mathbf{Y}})\|^2, \quad (6)$$

where $\odot$ is the element-wise product, $\mathbf{Y}$ is the DTI matrix with observed non-zero entries, $\|\cdot\|^2$ is the squared Frobenius norm and γ is a balance coefficient.

Result Dataset

In this work, we adopted the DTInet dataset compiled by the previous study [2]. As shown in Table 1, the DTInet dataset includes a DTI-related HN with six edge types: drug-protein interaction, drug-drug interaction, protein-protein interaction, drug-disease association, protein-disease association and drug-side effect association. For all drugs and proteins in the dataset, we collected SMILES of drugs from Drugbank [36] and sequences of proteins from Uniprot [37].

Experiment settings

In our experiments, we treat the drug-target pairs with known interactions as positive samples and the remaining drug-target pairs as negative samples. We randomly choose 10% of positive samples and a matching number of negative samples as the independent test set. The remaining samples are used as the cross-validation set. The cross-validation set is applied to 10-fold cross-validation, where the positive samples in the cross-validation set are equally divided into 10 portions, and in each fold, one portion of the positive samples, along with an equal number of negative samples randomly selected from the cross-validation set, is viewed as validation set, while the rest of samples in the cross-validation set serve as the training set. We consider several evaluation metrics: ROC-AUC (area under the receiver operating characteristic curve), PR-AUC (area under the precision-recall curve), F1 score, Accuracy, Recall. In each fold of the 10-fold cross-validation, we train the predictive models on the training set and calculate the metrics on the validation set. Doing so will yield 10 values for each metric and we use the average values as the 10-fold cross-validation results as well as calculate the standard deviations.

Table 2. Performances evaluated on the cross-validation set

	ROC-AUC	PR-AUC	F1	Accuracy	Recall
NetLapRLS	0.9074	0.9180	0.8388	0.8321	0.8730
CMF	0.9010	0.9247	0.8468	0.8487	0.8367
DTInet	0.8612	0.8861	0.7996	0.8029	0.7853
NeoDTI	0.9150	0.9173	0.8562	0.8550	0.8628
HampDTI	0.9273	0.9263	0.8689	0.8648	0.8986

The proposed method HampDTI has four key hyperparameters: (1) the embedding dimensionality F ; (2) the length of meta-paths contained in the constructed meta-path graph L ; (3) the number of channels C ; (4) the balance coefficient γ in Eq. (6). We consider all combinations from $F \in \{128, 256, 512\}$, $L \in \{2, 3, 4\}$, $C \in \{3, 4, 5\}$ and $\gamma \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$, and select the optimal one based on the 10-fold cross-validation results. Finally, we set $F = 128$, $L = 3$, $C = 4$, $\gamma = 0.4$. More implementation details the parameter analysis of HampDTI are provided in Supplementary Materials.

Comparison with baselines

Baselines

We compare the performance of HampDTI with the following: HN-based DTI prediction methods, including three classic shallow models, NetLapRLS [38], CMF [39] and DTInet [2]; and a state-of-the-art deep model, NeoDTI [10].

NetLapRLS is a manifold regularization semi-supervised learning method by integrating information from drug-protein interaction network, drug-drug similarity network and protein-protein similarity network.

CMF is a multi-similarity cooperative matrix decomposition algorithm, which maps drugs and proteins into a common low-dimensional space. It integrates more diverse similarity network information compared with NetLapRLS.

DTInet uses RWR (Random Walk with Restart) and DCA (Diffusion Component Analysis) algorithm to obtain a low-dimensional vector representation of drugs and proteins, and then adopts inductive matrix completion to predict DTIs.

NeoDTI is a recent deep model for predicting DTIs based on HN, which employs a GNN to learn topology-preserving node embeddings by reconstructing the edge weight of the HN.

We adopt the implementations in NetLapRLS and CMF from the publicly available code (<https://github.com/hansaimlim/wiZAN/tree/master/PyDTI>) with default parameter settings. For DTInet and NeoDTI, we use the implementations provided in literature [2, 10], where the embedding dimension of targets and drugs are set to 128 as the same as HampDTI and other parameters hold default.

Performances evaluated by 10-fold cross-validation

Here, we implement the 10-fold cross-validation to evaluate performances of our method HampDTI and baselines

on the cross-validation set. As shown in Table 2, HampDTI outperforms all baselines, improving 1.3% in terms of ROC-AUC over the second best method.

The DTInet dataset may contain homologous proteins and similar drugs, which may lead to overoptimistic performance of the model due to easy predictions. Following the similar assessment strategy in [2, 10] to mimic rougher prediction, we conduct four additional tests on cross-validation set: (1) DTIs with either similar drugs (drug chemical structure similarities >0.6) or homologous proteins (protein sequence similarities >0.4) are removed; (2) DTIs with drugs sharing similar drug interactions (Jaccard similarities >0.6) are removed; (3) DTIs with drugs sharing similar side-effects (Jaccard similarities >0.6) are removed; (4) DTIs with drugs or proteins sharing similar diseases (Jaccard similarities >0.6) are removed. The results of the above tests are shown in Figure 2 (B–E), respectively, and we also display the results without removing similarity in Figure 2 (A) for convenient comparison. It is obvious that almost all methods produce decreased performances after removing the homologous proteins or similar drugs in training data. Nevertheless, HampDTI still yields comparable or even better performances compared with other baseline methods, which demonstrates the robustness of HampDTI in rough prediction scenario.

The DTInet dataset contains a bias that some of drugs and proteins have only one interaction (such interactions are called unique DTIs). This bias can be captured by prediction methods to achieve an overoptimistic performance when evaluating those methods by 10-fold cross-validation [40]. To mitigate this issue, we follow the previous study [10] to evaluate the prediction performance of HampDTI by separating unique DTIs from non-unique ones. We train all methods on non-unique DTIs and then evaluate them on unique DTIs, which also mimics the scenario of predicting the DTIs of new drugs or targets that have few prior DTI knowledge. The results are shown in Figure 2 (F). We can see that HampDTI significantly outperforms all baseline methods, with 4.4% higher in terms of ROC-AUC than the second best method, which shows the decent inductive capability of HampDTI. Taken together, HampDTI has the great potential in predicting new DTIs for drugs or targets that do not have much prior DTI knowledge.

Performances evaluated by independent test

To further investigate the generalization performance of the predictive models, we evaluate the prediction ability

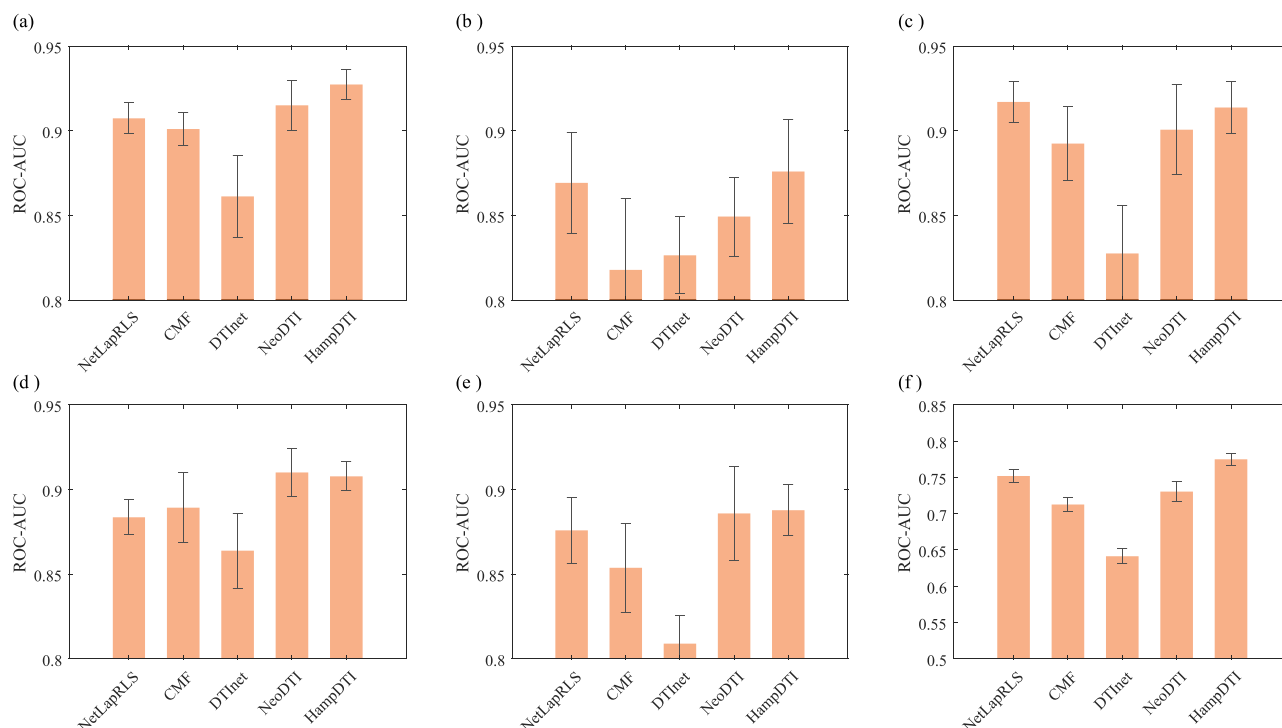


Figure 2. The performance evaluation results of HampDTI on several challenging scenarios in terms of ROC-AUC scores. In each scenario, we compare HampDTI with all the baseline methods in a 10-fold cross-validation test. (A) results on the cross-validation set. (B) results on the cross-validation set when removing DTIs with similar drugs or homologous proteins. (C) results on the cross-validation set when removing DTIs with drugs sharing similar drug interactions. (D) results on the cross-validation set when removing DTIs with drugs sharing similar side-effects. (E) results on the cross-validation set when removing DTIs with drugs or proteins sharing similar diseases. (F) results of models trained on non-unique interactions and tested on unique interactions.

Table 3. Performances evaluated on the independent test set

	ROC-AUC	PR-AUC	F1	Accuracy	Recall
NetLapRLS	0.9003	0.8993	0.8506	0.8515	0.8469
CMF	0.8970	0.9210	0.8461	0.8487	0.8313
DTInet	0.8505	0.8567	0.7928	0.7823	0.8333
NeoDTI	0.9057	0.9002	0.8518	0.8458	0.8839
HampDTI	0.9209	0.8972	0.8824	0.8776	0.9193

of them on the independent test set that is not used for model training and hyper-parameter determination. Since each method has been trained once in each fold of 10-fold cross-validation, we evaluate it on the independent test set. Thereby, for the independent test, we obtain 10 values for each kind of metric and use the average values as the results of the independent test. As shown in Table 3, HampDTI performs the best among them in terms of most metrics, especially improves 3.7% on average over other baseline methods in terms of ROC-AUC which accords with the results on the cross-validation set, indicating a good generalization capability for unseen data.

Ablation experiments

To examine the contribution of each component of HampDTI, we conduct an ablation study by considering several variants:

- **HampDTI-Feature:** we remove the molecule characterizer, and use randomly initialized vectors in GNN-based encoder instead.
- **HampDTI-Meta-path:** we remove the meta-path graph constructor and only use the DTI network in the GNN-based encoder.
- **HampDTI-DTI:** we remove the DTI network and only use the constructed C-channel meta-path graphs in the GNN-based encoder.
- **HampDTI-NAGNN:** we replace the devised node type-aware GNN with the vanilla GCN [30].

All prediction models are evaluated by 10-fold cross-validation on the cross validation set, and results are shown in Figure 3. In general, all variants produce the decreased performances, indicating all the components are essential to ensure the high-accuracy prediction of HampDTI. The great gap between HampDTI and HampDTI-Feature shows that the drug and protein features learned from drug molecules and protein

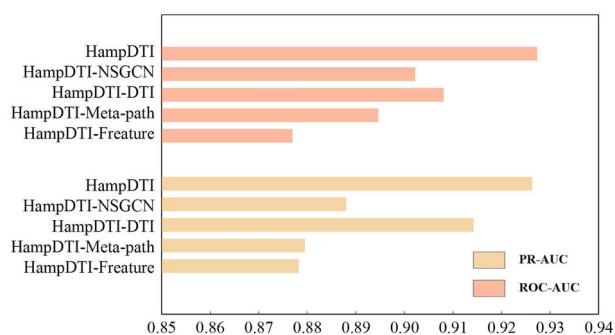


Figure 3. The result of the ablation study.

sequences can significantly enhance the performance compared with random initialization of node features in the GNN-based encoder. The results of HampDTI and HampDTI-Meta-path demonstrate that the constructed meta-path graphs can leverage the direct DTIs and the indirect relationships between drugs and targets which are beneficial for the DTI prediction. The results of HampDTI, HampDTI-Meta-path and HampDTI-DTI indicate that indirect associations in the HN are more informative than the direct ones. Compared with the vanilla GCN employed by HampDTI-NAGNN on the meta-path graphs, the devised node type-aware GNN adopted by HampDTI produces much better results which demonstrates our devised node type-aware GNN can capture more information to enhance the prediction performance.

Discussion

The main contribution of this paper is the design of the proposed model HampDTI to enable automatic extraction of important meta-paths from the relevant HN for DTI prediction. Automatically extracting meta-paths has two advantages. First, it avoids dependency on domain knowledge and can adaptively learn meta-paths between drugs and targets from the relevant HN. Second, it can reveal some of the most important meta-paths for DTI prediction and deliver some valuable insights to DTI-related studies, which provides better interpretability compared with the previous black-box deep learning models.

The identification of useful meta-paths comes from the well-trained relation-specific attention weights $\tilde{\alpha}_{i,r}$ in Eq. (2). As described in previous section, the $\tilde{\alpha}_{i,r}$ is a weight assigned to measure the importance of the relation r at i th step in one extracted meta-path. Hence, for each channel of the constructed meta-path graph, we can select the most important meta-path according to the trained attention weights. The selection scheme is illustrated in Figure 4. The augmented relation set used in Eq. (2) contains 11 relations composed of the relation identical (I) and the relations among four bio-entity types that are protein (pr), drug (dr), disease (di) and side effect (se). Figure 4 (A) visualizes one channel's

the attention weight $\tilde{\alpha}_{i,r}$ of the HampDTI trained in one fold of the 10-fold cross-validation experiment, where $\tilde{\alpha}_i \in [0, 1]$ ¹¹ is the attention vector at the i th step. The dashed lines show all four possible meta-paths if we select the relation $pr-pr$ (protein-protein interaction) at the first step and only consider drug as the ending node type.

Each possible meta-path is showed as a path linked by dash lines corresponding to a sequence of attention weights. We quantify the possibility of one meta-path by the meta-path score defined as the product of these weights on a path of dash lines. We can list all possible meta-paths between drug and protein, and accordingly calculate the meta-path scores. Further, the relative importance of one meta-path is measured by the ratio of the corresponding meta-path score to the sum of all meta-path scores. Figure 4 (B) shows all the possible meta-paths and the relative meta-path score.

We find that the top six important meta-paths reaches 97% contribution and some of them accord with this study [27] though the used datasets are different. Fu et al. [27] listed 51 meta-paths with length ranging 2 to 4 and compared the importance of them by random forest-based prediction performance, where six meta-paths shown in Figure 4 (B) are included in their meta-path list: $dr \rightarrow se \rightarrow dr \rightarrow pr$, $dr \rightarrow pr \rightarrow dr \rightarrow pr$, $dr \rightarrow di \rightarrow dr \rightarrow pr$, $dr \rightarrow pr \rightarrow pr$, $dr \rightarrow di \rightarrow pr$ and $dr \rightarrow pr \rightarrow di \rightarrow pr$, respectively, matching C10, C7, C11, C2, C4 and C15 in their meta-path list. Among these meta-paths, the ranking of C7 that is our top-four are considerably high (top-13 in 51) and C10 (our top-one) ranks in the middle in their study. Compared with the other three meta-paths, C2, C4 and C15 are in the bottom of the ranked list in their study, as the same finding in this study (ranked in the bottom six). Interestingly, we find that the meta-path $dr \rightarrow pr \rightarrow pr \rightarrow pr$ (top-six) is relatively important for DTI prediction, which is valid for their dataset but not considered in their study, indicating our proposed HampDTI has the potential to discover underlying knowledge from complex heterogeneous graphs.

Conclusion

In this work, we propose a novel HN-based method with attentive meta-path extraction for DTI prediction, named HampDTI. HampDTI can extract meta-paths helpful for DTI prediction, which avoids the dependence on domain knowledge and provides new perspectives for HN-based DTI prediction. The experimental results show that HampDTI outperforms some baseline methods, and demonstrate the practicability of HampDTI discovering important meta-paths. In future work, we will design effective fusion mechanism to merge features of multiple bio-entities and extend the devised models to other HN-based link prediction tasks, such as drug-disease associations [41, 42] and protein-protein interactions [43, 44].

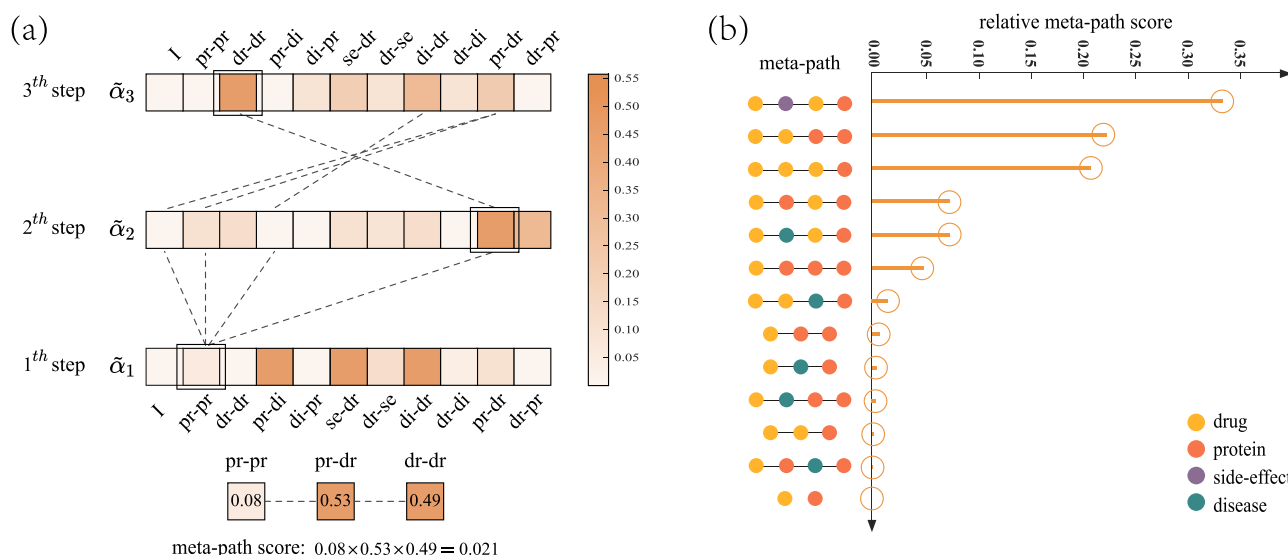


Figure 4. Visual presentation of the meta-paths learned by HampDTI. (A) Visualization of the attention vector $\tilde{\alpha}_i \in [0, 1]^{11}$ at the i th step in meta-path graph constructor. (B) The analysis of relative meta-path scores among all 13 meta-paths in the DTInet heterogeneous network.

Key Points

- We design a novel heterogeneous network-based method for predicting drug-target interactions, termed HampDTI.
- HampDTI can adaptively extract valuable meta-path information for DTI prediction without any pre-defined meta-paths, which provides a new perspective for interpretability on how drugs interact with targets.
- we devise a node-type aware graph neural network to integrate drug structure features and target sequence features as well as the extracted meta-path information.
- The experimental results show that HampDTI can outperform several baseline methods and demonstrate its capability of extracting useful meta-paths.

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Supplementary Data

Supplementary data are available online at....

Author contributions statement

W.Z., H.W. and F.H. conceived the experiment(s), H.W. conducted the experiment(s), Z.X. analyzed the results. H.W. and F.H. wrote the manuscript. All authors reviewed the manuscript.

Data availability

The data sets and source code can be freely downloaded from <https://github.com/MacroHongZ/HampDTI>.

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